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Review Article

The Beneficial Role of Early-Life Gut Microbiota Development for Improving Intestinal Health Integrity and Gut-Immune Function in Infants of Bangladesh

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Abstract

Early life represents a critical window for the establishment of the gut microbiome, a complex microbial ecosystem that plays a central role in intestinal health, immune system development, physical growth, and long-term disease risk. In low- and middle-income countries such as Bangladesh, infants are exposed to a range of nutritional and environmental stressors—including undernutrition, recurrent infections, and inadequate sanitation—that can significantly disrupt normal microbiome development. These challenges make the study of early-life microbiome assembly particularly important in this context. This review synthesizes recent research on gut microbiome development in Bangladeshi infants and explores how deviations from normal patterns—such as microbiome immaturity, dysbiosis, and the overrepresentation of pathogenic organisms—are associated with adverse outcomes. These include impaired intestinal function, chronic inflammation, weakened immune responses, and growth faltering. Studies consistently show that healthy infants typically undergo a predictable succession of microbial colonization, with beneficial taxa such as Bifidobacterium infantis and other members of the Bifidobacterium longum group dominating early life, particularly in breastfed infants. These microbes are crucial for metabolizing human milk oligosaccharides and supporting immune maturation. In contrast, malnourished infants often exhibit delayed or altered microbiome maturation. Their gut microbial communities resemble those of younger infants, indicating immaturity, and are often enriched with harmful or less beneficial bacteria. Importantly, conventional nutritional interventions alone have been shown to only partially restore a healthy microbiome composition, suggesting that more targeted approaches are necessary. Recent advances highlight the potential of microbiota-directed complementary foods (MDCF) and specific probiotic supplementation in addressing these issues. These interventions are designed to promote the growth of beneficial microbes and restore microbial balance. Evidence indicates that MDCF can accelerate microbiome repair, enhance gut barrier integrity, reduce intestinal inflammation, and contribute to improved growth outcomes in undernourished children. In conclusion, targeting gut microbiome development during early infancy offers a promising strategy to improve health outcomes among Bangladeshi children. However, further research is needed to determine the optimal timing, duration, and sustainability of such interventions, as well as to better understand the underlying biological mechanisms driving these benefits.

Keywords

Infants Intestinal Health, Gut Microbiome, Gut Immunity, Malnutrition, *Bifidobacterium*, Microbiota-directed Complementary Foods, Environmental Enteric Dysfunction

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